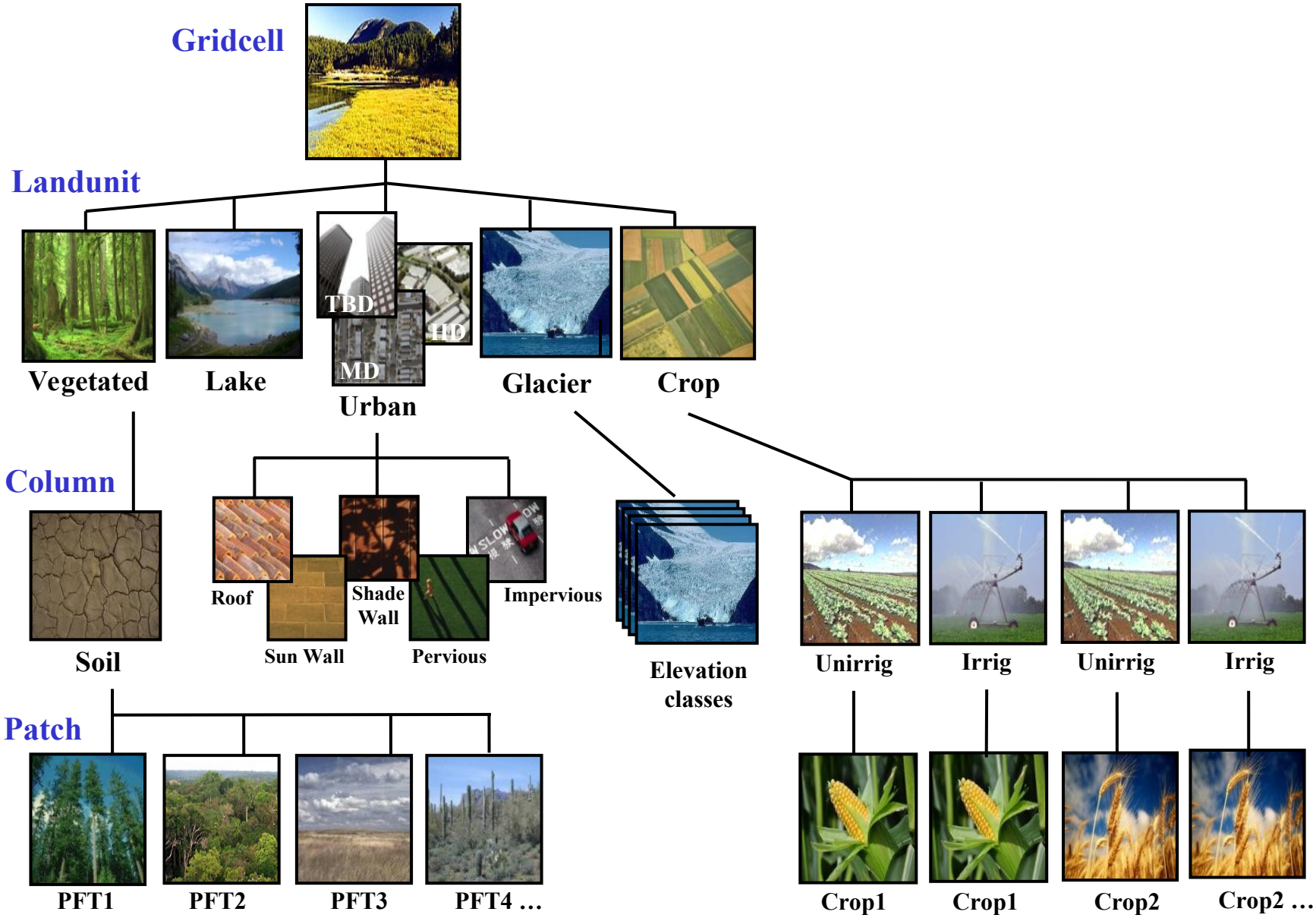
Hillslope Postprocessing

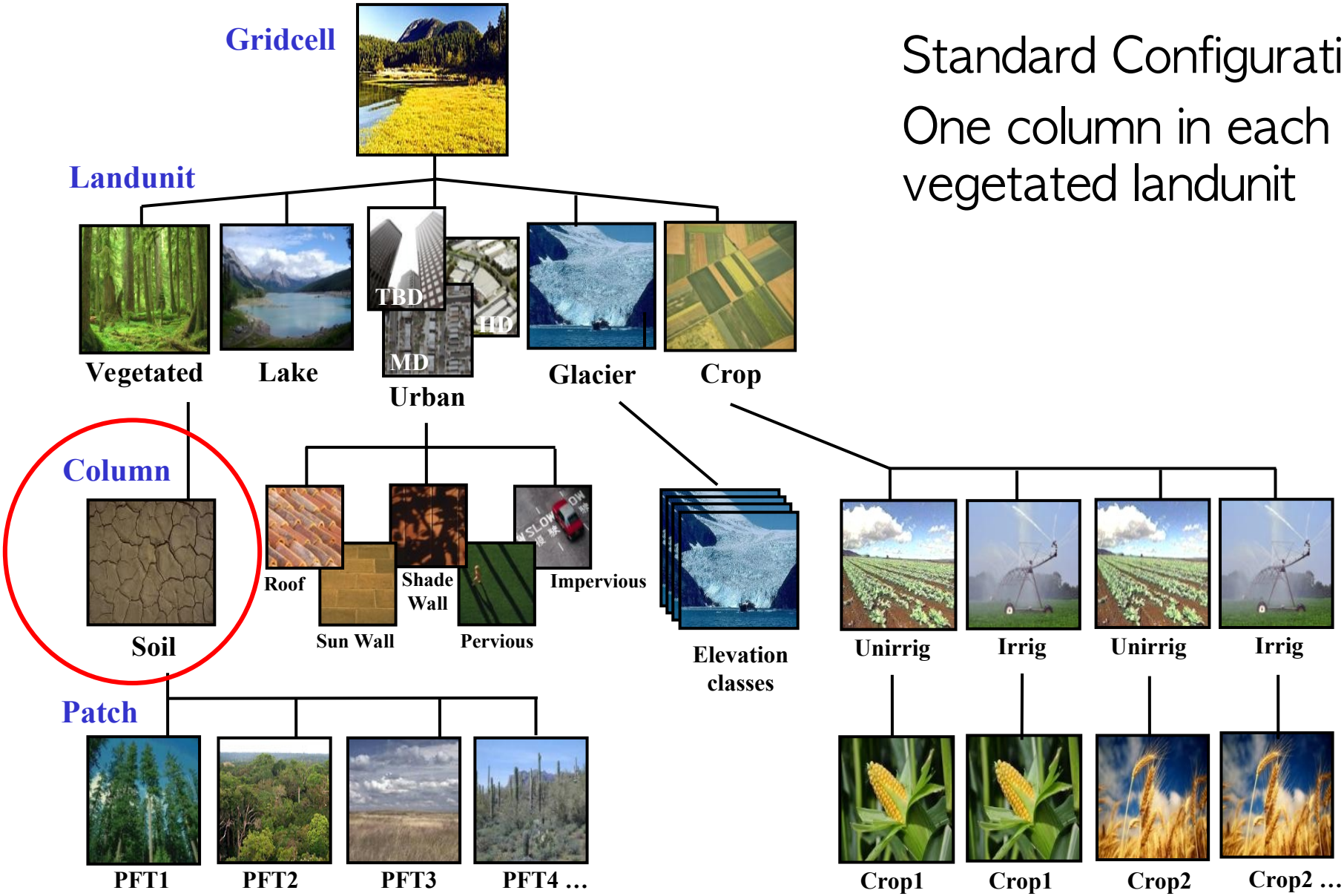
Individual hillslope column states and fluxes are saved to vector history files

To visualize column-level data (i.e. to create maps), it must be converted from vector (1d) to gridded (2d) format

CTSM Subgrid Tiling Structure



CTSM Subgrid Tiling Structure



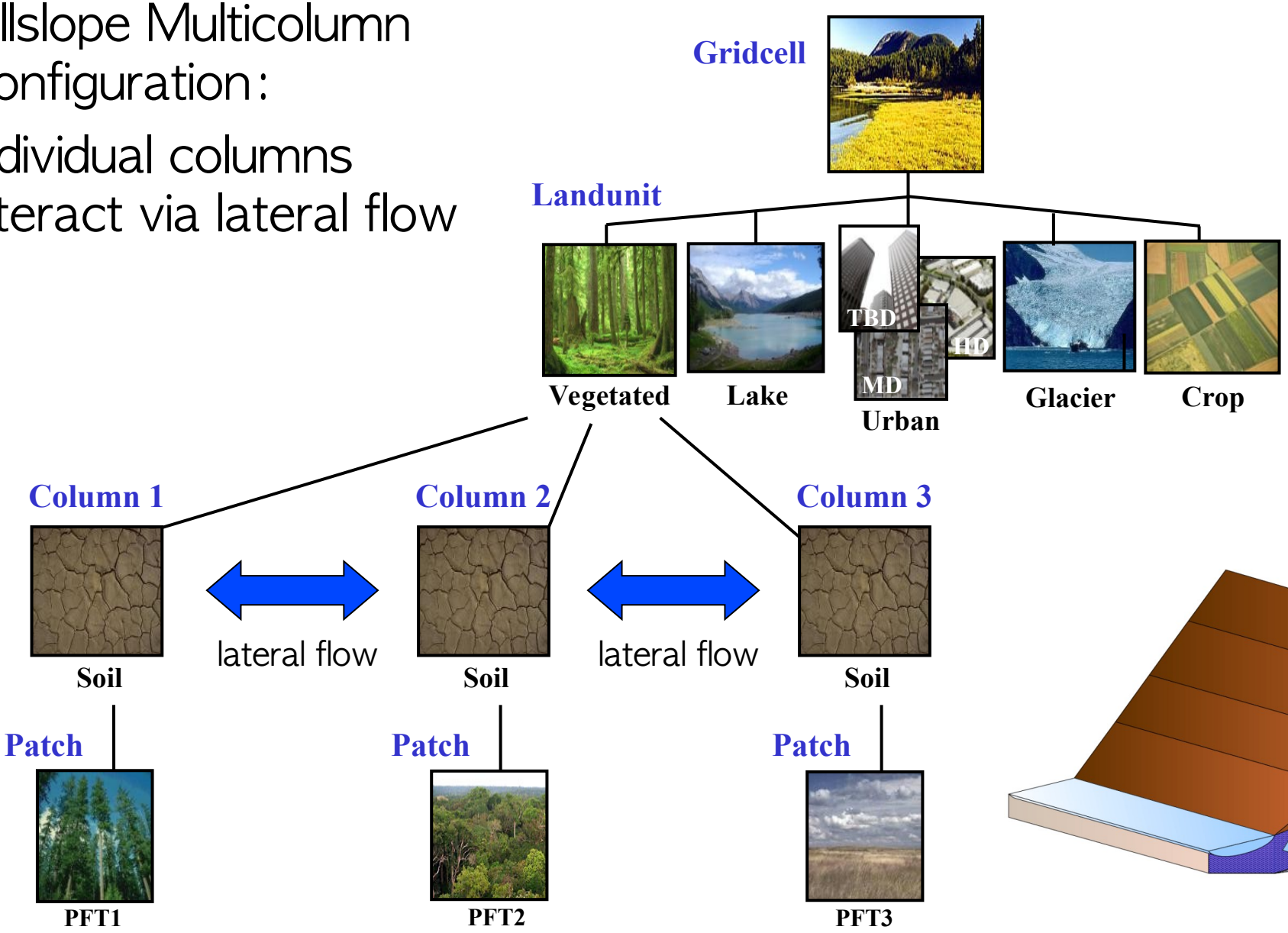
Standard Configuration:

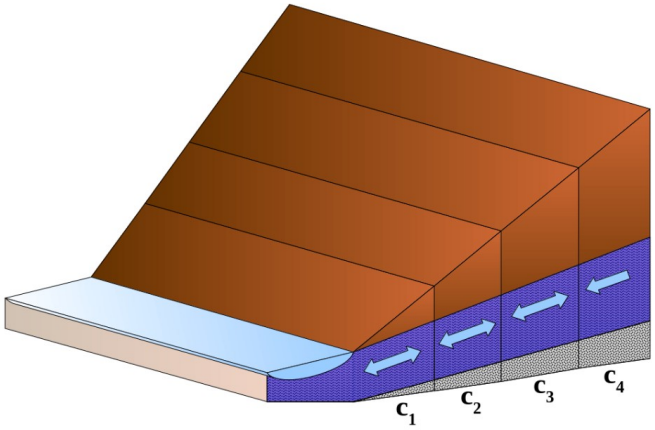
One column in each vegetated landunit

CTSM Subgrid Tiling Structure

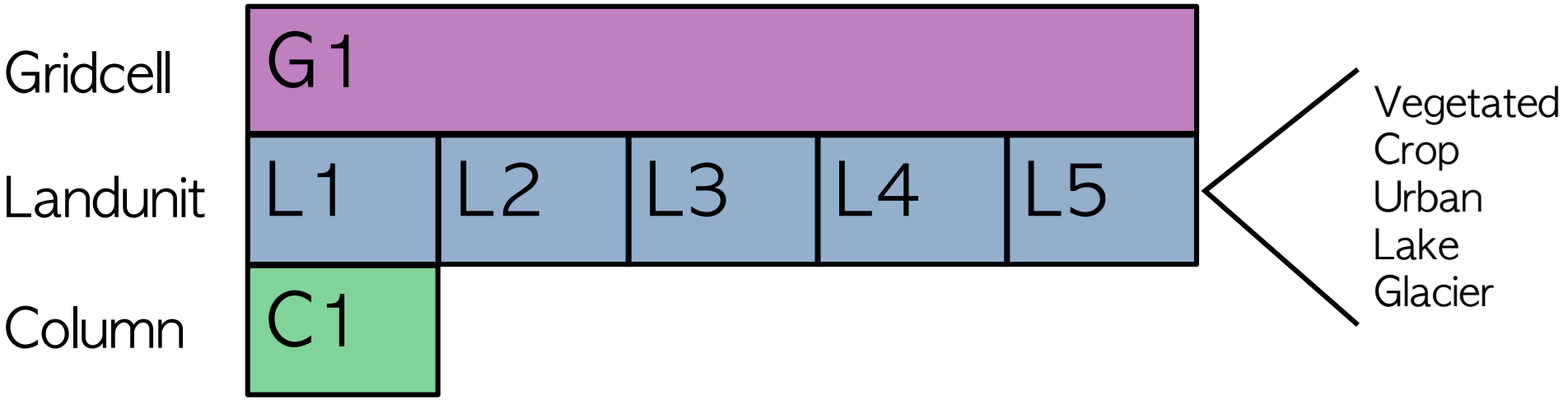
Hillslope Multicolumn Configuration:

Individual columns interact via lateral flow



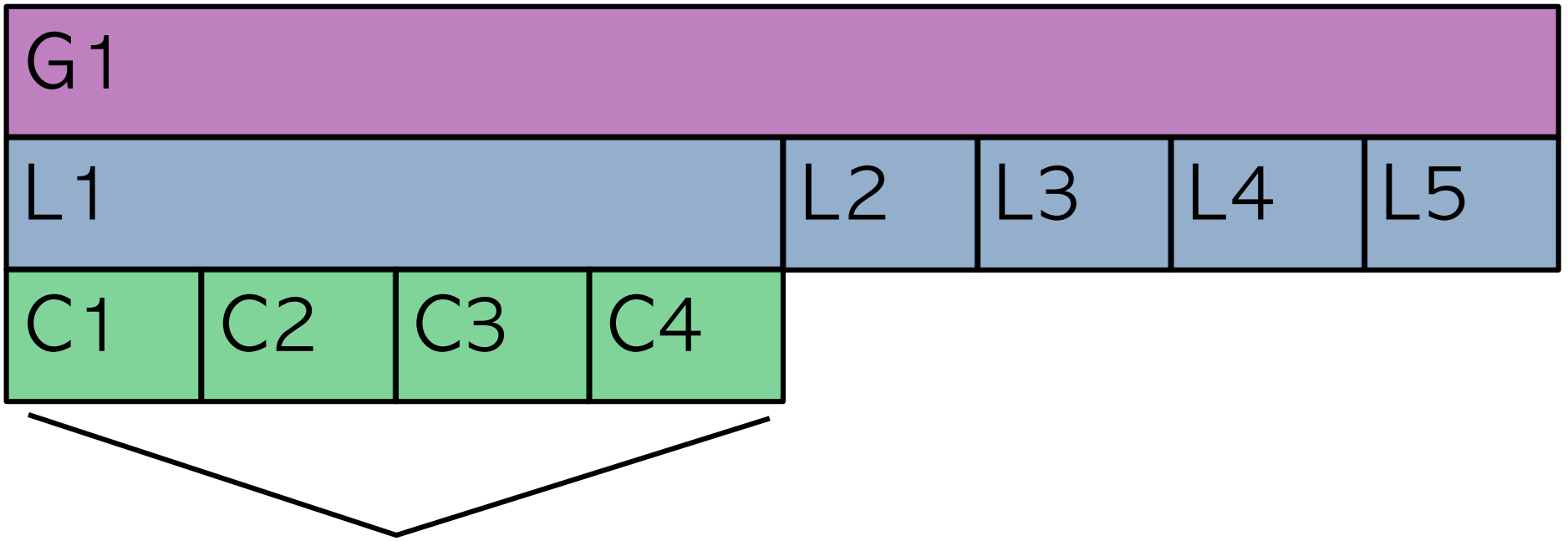


CTSM Vector Output



In the standard configuration, there is a 1-to-1 mapping from column to gridcell, so visualizing column-level variables is straightforward

CTSM Vector Output



In this example, there are four hillslope columns in G1, each having the same latitude and longitude

In the hillslope configuration, the mapping from column to gridcell is degenerate, so visualizing column-level variables is less straightforward

CTSM Vector Output

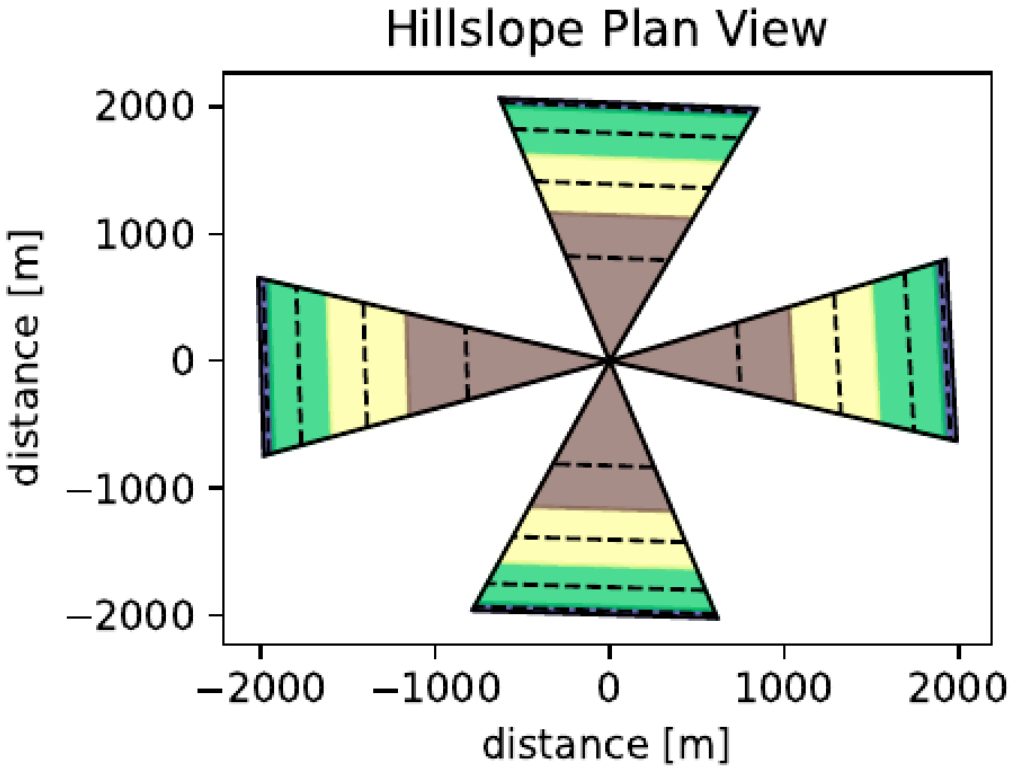
Vector-to-gridded workflow:
From the vector output, a single column per gridcell can be plotted or columns can be aggregated.

For example:

Single column: north, lowland

Single hillslope:
mean of all north columns
(Requires area-weighting)

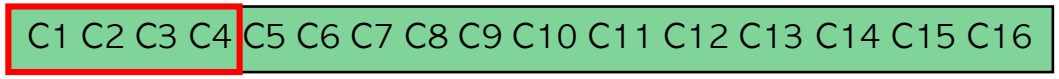
All hillslope columns:
(exclude other landunits)



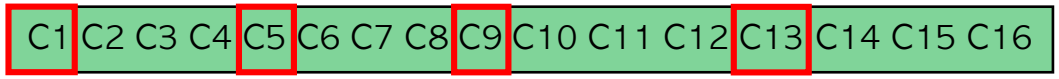
Each triangle in plan view represents a hillslope, and each color is a different column. In this example, there are 16 columns.

CTSM Vector Output

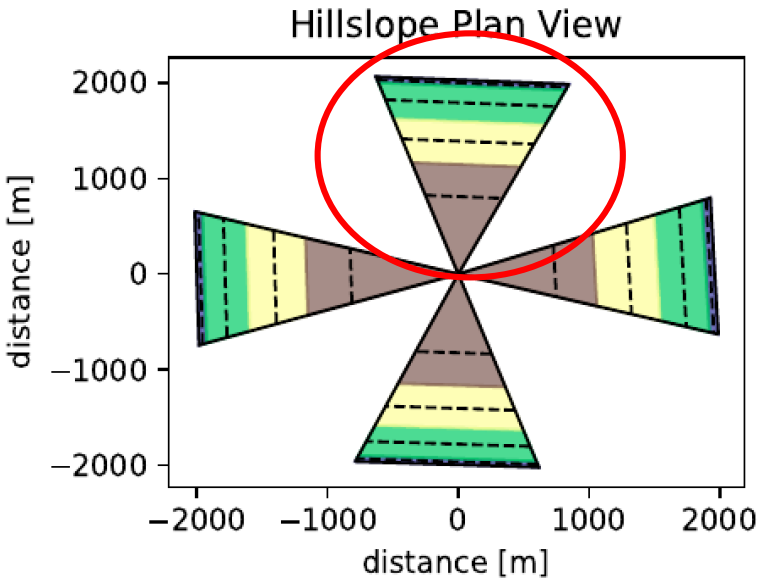
Columns are ordered by hillslope (N,E,S,W), in order of increasing elevation

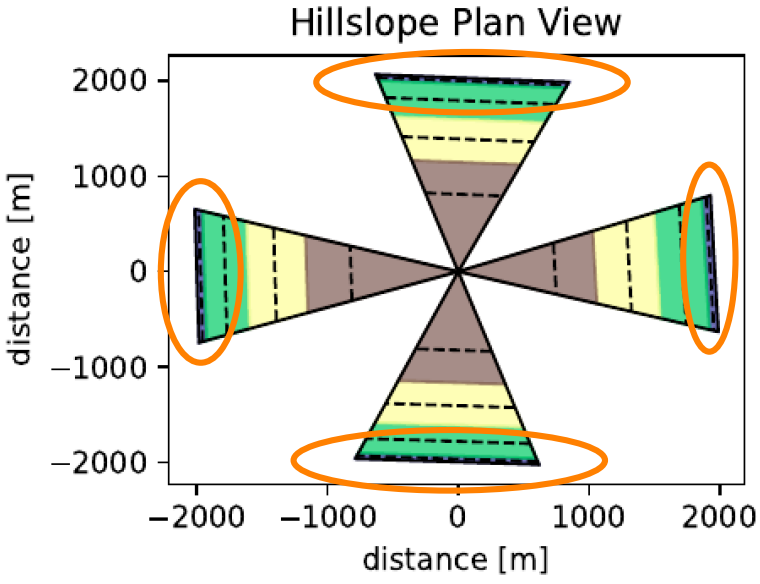


These are the columns in hillslope 1 (North-facing)



These are the lowland columns in each hillslope





Hillslope Postprocessing

1) convert all vector history files to gridded files for each subgrid unit (e.g. column or patch/pft)

2) convert vector to gridded as needed without saving new files

Hillslope Postprocessing

1) convert all vector history files to gridded files for each subgrid unit (e.g. column or patch/pft)

Pro: "one-and-done", all data available for use in standard visualization and analysis workflows

Con: large number of additional files with redundant information

Hillslope Postprocessing

2) convert vector to gridded as needed without saving new files

Pro: no storage overhead, less time required

Often useful when looking at a single gridcell

Con: may slow down workflow